

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech. Examination (EC)****March-Apr 2018****EC 304.01/304 Power Electronics****Date: 03/04/2018, Tuesday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70*****Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1** Why do we need UPS? Explain the operation of different architecture of UPS with necessary diagrams? [07]
- Q - 2 (a)** Explain the ON OFF controller for ac to ac conversion with the necessary diagrams and equations [07]
- (b)** Explain in brief, the following term with respect to thyristor. [07]
- 1) Latching Current
 - 2) Holding Current
 - 3) Turn off and Turn on time

OR

- Q - 2 (a)** Explain the principle of single phase semi converter operation. [07]
- (b)** What are the different methods for turning off an SCR? Explain Class C commutation with necessary circuit diagram and waveform in detail. [07]
- Q - 3 (a)** Explain single phase Half controlled bridge rectifier with appropriate waveform and diagrams. [10]
- (b)** Explain the Control Characteristics of Different Power Devices. [04]

OR

- Q - 3 (a)** Explain in detail the Buck Boost Regulator? [10]
- (b)** Write a short note on: (i) SCR (ii) IGBT [04]

SECTION – II

- Q - 4** How does di/dt and dv/dt affect the power semiconductor devices? How can you protect your devices from it? [07]
- Q - 5 (a)** Derive the equation for initial rate of change of capacitor voltage for diodes with RC load. [07]
- (b)** Write in brief about different operating modes of DC motors with necessary diagrams. [07]

OR

Q - 5 (a) Explain the principle of operation of single phase bridge inverter with necessary equations of output rms voltage, instantaneous output voltage, instantaneous load current and fundamental power. **[07]**

(b) Explain the flyback converter with necessary diagram and equations. **[07]**

Q - 6 (a) Explain three phase half wave controlled rectifier with necessary diagram and waveforms. **[10]**

(b) Define the following terms related to controlled rectifier. **[04]**

- 1) Rectification efficiency
- 2) Form factor
- 3) Ripple factor
- 4) Transformer utilization factor.

OR

Q - 6 (a) Draw the circuit of three-phase inverter and plot all phase voltages and line voltages for 180 degree mode. **[10]**

(b) Write down the characteristics of an ideal Switch. **[04]**
